4.00 USD

1. **Sentinel Smart Valve Actuator (Model S-WFC01)**

- *Based on Ecowitt WFC01*
- Smart water timer valve designed to mitigate localized wildfire threat by auto-wetting property grounds when dry soil moisture thresholds trigger parametric alarm states.
- **Price:** $79.00 USD

---

2. Automated Network Onboarding Workflow

The Shopify purchase acts as the initial transaction in the node onboarding lifecycle.

~~~
``mermaid

sequenceDiagram

autonumber

actor User as Policyholder / Buyer

participant Shop as Shopify Storefront

participant Webhook as Shopify Webhook Worker

participant API as Chainmail API Server

participant DB as Firestore Database

participant Device as IoT Sentinel Device

User->>Shop: Purchase Sentinel Kit (S-100 + S-700)

Shop-->>User: Order Confirmation & Shipping Label Generated

Shop->>Webhook: Event: order/paid (Payload contains email, address)

Webhook->>API: POST /api/onboarding/provision

activate API
~~~

API->>API: Generate secure device key & pre-provision propertyId

API->>DB: Write property & pre-provisioned sensors as PENDING_ACTIVATION

API-->>Webhook: Provisioning Success

deactivate API

Webhook-->>User: Onboarding Email with Device Credentials & Claim QR Code

User->>Device: Mount hardware & connect to home Wi-Fi

Device->>API: POST /api/ingest/ecowitt (First Telemetry Ping)

activate API

API->>API: Authenticate device with secure pre-provisioned key

API->>DB: Update status to VERIFIED_NODE & Activate Parametric Coverage

API-->>Device: Acknowledge (HTTP 200)

deactivate API

API->>User: App Push Notification: "Node Active. Coverage Live!"

`

---

3. API Provisioning Endpoint Specifications

When a checkout completes, Shopify sends an `order/paid` webhook to our backend API server:

Endpoint: **POST /api/onboarding/provision**

Access Control: HMAC Signature verified using Shopify API Webhook Shared Secret.

Request Payload (from Shopify):

`json

{

"order_id": 987654321,

```
"email": "customer@example.com",

"shipping_address": {

"address1": "12665 Julington Oaks Dr",

"city": "Jacksonville",

"province": "FL",

"zip": "32223",

"country": "US"

},

"line_items": [

{

"sku": "SENTINEL-KIT-A",

"quantity": 1,

"name": "Sentinel Gateway & Wittboy Array"

}

]

}
```

`

Provisioning Logic:

1. **Address Validation:** Call Google Address Validation API to verify CASS status and retrieve exact rooftop coordinates (lat / lng).
2. **Database Record Creation:**

- Create a new document in /properties:

`typescript

{

ownerEmail: "customer@example.com",

address: "12665 Julington Oaks Dr, Jacksonville, FL 32223",

lat: 30.1421936,

lng: -81.6157784,

riskScore: 72.4, // Calculated by Underwriting Agent based on coordinates

status: "PENDING_ACTIVATION",

createdAt: serverTimestamp()

}

`

- Create a document in /properties/{propertyId}/sensors/Anemometry:

`typescript

{

type: "Anemometry",

verificationStatus: "UNVERIFIED",

hardwareType: "Ecowitt",

value: 0

}

`

1. **Secret Generation:** Generate a unique password/secret token for the gateway and email it as a QR code to the user.

---

4. UI/UX Storefront Integration

The Shopify storefront theme will match Chainmail's core brand design system: **Cobalt & Gold / Slate & Gold**.

Theme Design System

- **Primary Background:** #030712 (Slate Black)
- **Secondary Cards:** rgba(17, 24, 39, 0.7) (Translucent Dark Blue)
- **Accents:** #D4AF37 (Metallic Gold)
- **Fonts:** Outfit (Headings) and Inter (Body)

Custom Liquid Elements

- **Network Node Counter:** A live ticker in the header showing: ACTIVE_NODES: 1,482 | TOTAL_COVERAGE_LOCKED: $34,891,000`.
- **Map Twin Preview:** An interactive WebGL globe on the landing page showing glowing gold dots representing active coverage sensors in real time.
- **onboarding-tab widget:** Built-in section on the product page showing the simple "Plug, Scan, Insure" 3-step setup guide.

,500 payout cap) just so we can get their sensor data and help protect the community.
The Claims Adjuster Agent: When a disaster hits, this AI looks at satellite photos to verify the damage. If it's confident, it pays the customer instantly. If it isn't sure, it sends the claim to a human to review—it is strictly programmed *never* to automatically deny a claim.

The Vigilance Oracle Agent: This AI constantly watches the weather channel. If it sees a Category 5 hurricane coming, it flags the map and readies our bank accounts to pay people before they even ask.

The Quant Brain Actuary Agent: This AI watches our bank account. If we take on too much risk in one state, it automatically buys "reinsurance" (backup insurance for us) to keep the company financially safe.

Your Day-to-Day Responsibilities

As a Product Owner, you will:

- **Monitor the Vision:** Look at the Enterprise Dashboard. Are the AI agents making decisions that align with the BallPark Initiative?
- **Prioritize the Backlog:** Work with the Product Manager to decide which neighborhood of our "city" needs improvement next.
- **Understand the Costs:** The Enterprise dashboard shows you exactly how many fractions of a cent the AI is spending to make decisions. You balance the cost of the AI against the value it provides.

You don't need to know how to code the AI. You just need to know what the AI *should* be doing to help our customers. Welcome to the team!

A Learning & Development (L&D) Training Manual

Welcome to the Future of Insurance!

Welcome to Chainmail Solutions! Whether you are a Customer Service Rep, an Enterprise Administrator, or a Claims Reviewer, this guide will show you how to use the Chainmail software.

You don't need to be a computer programmer to use our system. The software does the heavy lifting for you!

What is Chainmail?

Traditional insurance is slow. If a hurricane hits, people wait months for an adjuster to look at their roof.

At Chainmail, we use **Parametric Insurance**. This means we have sensors (like weather stations) all over the country. If the sensor says the wind blew 150 MPH, we just pay the customer instantly. No waiting, no arguments. We do this to support our community (we call this the BallPark Initiative).

How to Use the System (Your Role)

If you are an Enterprise Administrator

You will spend your time in the **Enterprise Dashboard**.

1. **The Map:** You will see a live map of the United States. Green dots are safe customers, red dots are customers in danger.
2. **The AI Audit Tab:** We have four "Robots" (AI Agents) that work for us in the background. They approve policies and check claims automatically. On this tab, you can watch exactly what the robots are doing in real-time. If you see a robot making a mistake, you can report it to your Product Manager.

If you are a Claims Reviewer

Most of the time, our **AI Claims Adjuster** will approve payouts instantly without needing your help. However, if the AI is confused—say a user uploads a blurry photo, or the weather sensors don't match the customer's story—the AI will route the claim to YOU.

1. You will log into the dashboard and click the **Review Queue**.
2. You will see the AI's notes (e.g., *"I am routing this to a human because the photo looks manipulated"*).
3. You will review the evidence manually. **Remember our philosophy:** We want to find ways to cover people, not look for excuses to deny them. If it's a valid claim, hit **Approve**.

The "IoT Baseline" Feature (For Customer Support)

Sometimes, a customer will call and say, *"I tried to buy a $500,000 policy but it only offered me ,500."*

Here is how you handle it:

- Explain that because their house is in an **Extreme Risk Zone** (like right on the beach), our Underwriting AI cannot offer them half a million dollars right now.
- However, explain that we *did not reject them*. We offered them the **IoT Baseline Policy**. This gives them ,500 of emergency cash, and in return, they install a weather sensor on their house. This helps us gather data to protect the rest of their neighborhood!

Need Help?

If the screen turns white, or the map stops loading, do not panic! Just take a screenshot and send it to your Product Manager with the Jira ticket label `UI-BUG`.

Welcome to Chainmail! We are glad you're here.

,500."*

Here is how you handle it:

- Explain that because their house is in an **Extreme Risk Zone** (like right on the beach), our Underwriting AI cannot offer them half a million dollars right now.
- However, explain that we *did not reject them*. We offered them the **IoT Baseline Policy**. This gives them ,500 of emergency cash, and in return, they install a weather sensor on their house. This helps us gather data to protect the rest of their neighborhood!

Need Help?

If the screen turns white, or the map stops loading, do not panic! Just take a screenshot and send it to your Product Manager with the Jira ticket label UI-BUG.

Welcome to Chainmail! We are glad you're here.

5). Require D-U-N-S number for corporate verification. **TestFlight / Internal Testing** T-4 Weeks Release to internal stakeholders via Apple TestFlight and Google Play Internal Testing. **Store Submission** T-2 Weeks Submit to Apple App Review and Google Play App Review. (Buffer time for initial rejection and resubmission). **Public Launch** T-0 App is live, indexed, and available for download.

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2. Critical Apple App Store Requirements (iOS)

Apple is notoriously strict regarding financial and insurance applications. To pass the App Review Board, we must guarantee the following:

> WARNING
>
> **digital settlement & Financial Regulations (Guideline 3.1.5)**
>
> Because Chainmail utilizes automated execution rules on Secure L2 Ledger L3 for payouts using $USD-denominated digital settlement assets, the app *must* be submitted by a legal entity (not an individual developer) that is registered as a financial institution or has the appropriate licenses in the regions it operates.
>
> - **Action Item:** Ensure our corporate D-U-N-S number is used for the Apple Developer Account, and our legal entity name exactly matches the app publisher name.

> IMPORTANT
>
> **Data Privacy & Account Deletion (Guideline 5.1.1)**
>
> Apple absolutely mandates that any app allowing account creation *must* also provide an in-app option to completely delete the account and all associated data.
>
> - **Action Item:** Build an automated "Delete Account" API route in the Vercel backend.

- **App Tracking Transparency (ATT):** If we use analytics that track across third-party apps, we must implement the ATT prompt.
- **Minimum Functionality:** The app cannot just be a "wrapped website" (Guideline 4.2). It must utilize native iOS features (e.g., Push Notifications, FaceID/TouchID for secure login, local caching).

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3. Critical Google Play Requirements (Android)

Google Play has recently tightened restrictions on testing and financial apps.

> WARNING
>
> **20-Tester Rule**
>
> New Google Play developer accounts must run a closed test with a minimum of 20 opted-in testers for at least 14 continuous days before you can apply for production access.
>
> - **Action Item:** We must recruit 20 beta testers internally or via the Vigilance Grid network well before launch.

> IMPORTANT

> **Financial Features Policy**
>
> Google requires a prominent "Financial Features Declaration" during submission. If the app facilitates insurance or digital settlement payouts, we must provide links to our regulatory compliance documentation and operating licenses in our Google Play Console declaration.

- **Data Safety Section:** We must accurately declare exactly what data is collected (e.g., location data from Vigilance Grid IoT integration, financial info) and whether it is encrypted in transit.

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4. Next Steps for the Development Team

To transition our current Vercel web app into a compliant Native App:

1. **Architecture:** We must decide whether to wrap the Vercel app using a framework like Capacitor/Ionic, or rebuild the frontend natively using React Native.
2. **Privacy Policy URL:** A live, hosted Privacy Policy must be created on the public-facing website.
3. **Support URL:** A dedicated support and contact page must be established for user inquiries.

0/mo) post-funding for team access. **Database & Scaling** GCP Firestore / Blaze Plan 00.00 Scales dynamically; covers production compute and database queries. **Object Storage (Media/PII)** Cloudflare R2 $0.00 Selected specifically to eliminate AWS/GCP data egress fees. **Geospatial & Convective Mapping** XWeather API & Vexcel Imagery $750.00 Commercial subscription for severe convective storm mapping and high-res post-storm imagery. **Core AI Routing & HITL Verification** Google Gemini 1.5 Flash & 3.1 Pro $65.00 Production Gemini API tokens for IoT routing and human-in-the-loop validation. **Claims Fraud Audit** ISO Verisk ClaimSearch 50.00 Active double-claim checking API subscription. **Secure Registry & Smart Contract execution** Secure L2 Ledger Mainnet 5.00 Sponsored gas for transaction settlement. **Version Control** GitHub (Free Tier) $0.00 Upgrade to GitHub Team ($4/user) when engineering team expands. **Executive Compensation** COO Phase 1 (1099) $0.00 (Equity/$RMOR only) Transition to Full-Time Phase 1 (Salary + 8% Match) post-funding. **Total Monthly Burn ,090.00**

Future Capital Expenditure (Post-Funding)

Once institutional seed funding is secured, the following immediate upgrades will be unlocked to mature the ecosystem:

1. **CFAN Proprietary Licensing:** We will deprecate our reliance on open-source NWS/NCEP data and execute a commercial licensing agreement with the **Climate Forecast Applications Network (CFAN)**. This will drop our 5-day track error to ~131 miles, providing institutional credibility to reinsurers. *(Estimated: Negotiated Enterprise Contract)*

2. **Amazon IP Accelerator:** We will abandon the standard USPTO pending application process and utilize Amazon's vetted legal firms to fast-track our Sentinel IoT hardware Trademark and Brand Registry. *(Estimated: ,500 - $3,000)*
3. **DVN Node Infrastructure:** Funding the initial liquidity pools on Secure L2 Ledger to incentivize legacy Third-Party Vendors (TPVs) to transition into distributed Validator Network (DVN) node operators.